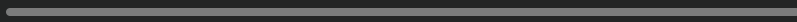




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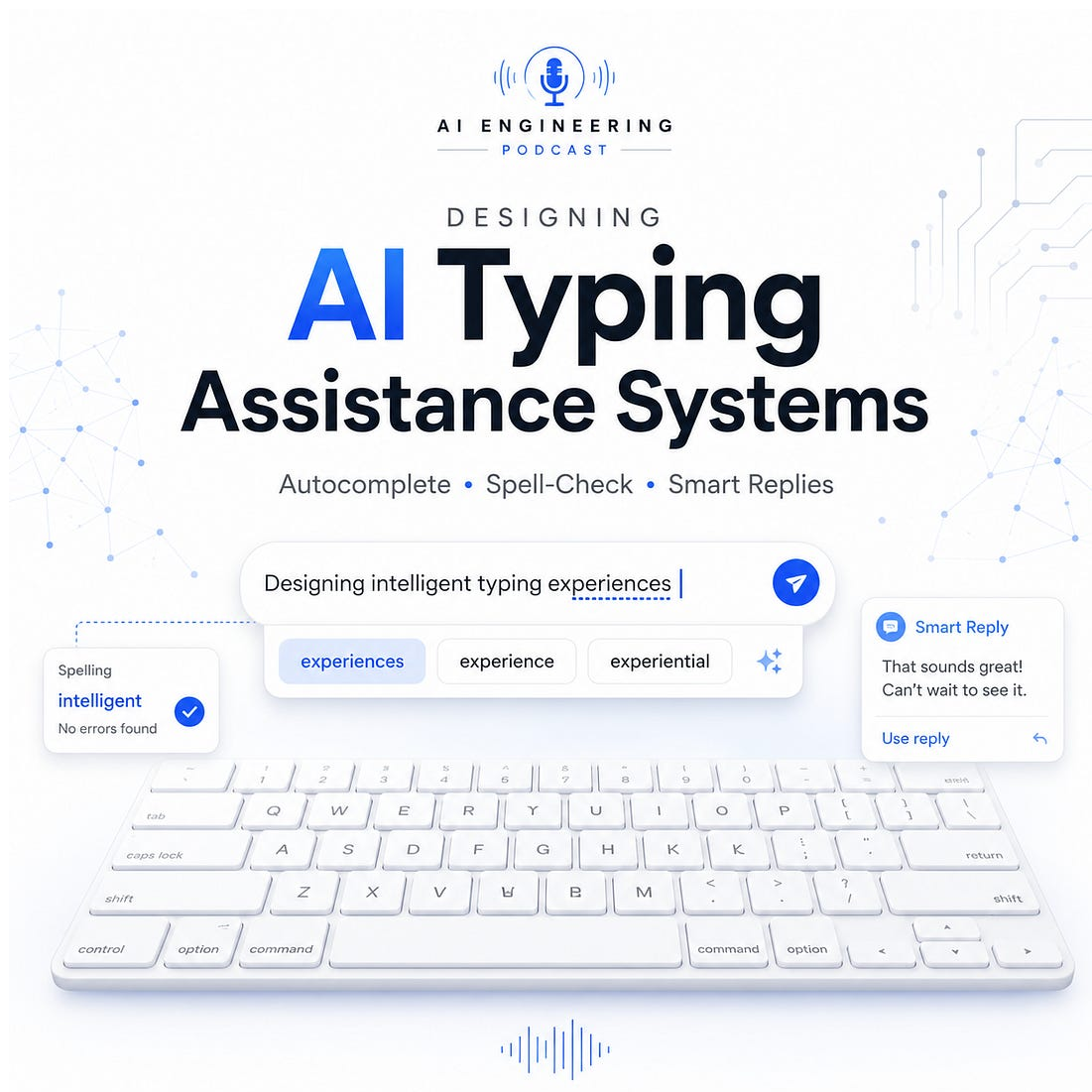
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Designing AI Typing Assistance Systems

How to Build Autocomplete, Spell-Check, and Smart Replies at Production Scale



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Typing assistance systems power some of the most frequently used AI features in modern products: autocomplete in search bars, spell-check in documents, smart replies in email, and next-word prediction on mobile keyboards. At first glance, these systems look simple: predict what the user wants to type next. In production, however, the challenge is much harder. The system must be fast, private, personalized, safe, multilingual, and accurate under extreme latency constraints.

For an AI system design interview, the key is to frame typing assistance as a **real-time NLP prediction system** optimized for user productivity.

The goal is not only to generate fluent text, but to reduce user effort while preserving trust.

1. Product Goal and Core Requirements

The main objective is to help users type faster and with fewer errors.

Examples include:

- ↳ **Autocomplete**: suggest the next word, phrase, or sentence.
- ↳ **Spell-check**: detect and correct misspelled words.
- ↳ **Grammar assistance**: improve sentence structure.
- ↳ **Smart replies**: generate short response options.
- ↳ **Personalized suggestions**: adapt to user vocabulary, tone, and repeated phrases.

The most important system requirement is **ultra-low latency**. Typing assistance must feel instantaneous. A strong target is under **50 milliseconds** for common suggestions, especially for keyboard and search autocomplete. If the system is slow, the user will ignore it.

Other requirements include:

- ↳ High relevance and contextual accuracy.
 - ↳ Strong privacy guarantees for sensitive text.
 - ↳ Personalization without exposing private user data.
 - ↳ Safety filtering to prevent offensive or harmful suggestions.
 - ↳ Support for multiple languages, domains, and devices.
 - ↳ Graceful degradation when network or model services fail.
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2. High-Level Architecture

A production typing assistant usually uses a **hybrid architecture**:

- ↳ On-device models for speed and privacy.
- ↳ Server-side models for heavier ranking and large language understanding.
- ↳ Local caches for frequent predictions.
- ↳ Federated learning for privacy-preserving personalization.

A common design pattern is **retrieve-then-rank**.

First, the system quickly retrieves candidate suggestions using lightweight methods. Then, a ranking model scores the candidates based on context, user history, and quality signals.

The architecture can be divided into five layers:

1. **Input Processing Layer**

Captures the current text prefix, cursor position, previous words, language, device context, and app context.

2. **Candidate Generation Layer**

Produces possible completions using prefix tries, dictionaries, n-gram language models, recent user phrases, and neural language models.

3. **Ranking Layer**

Scores suggestions using contextual relevance, probability, personalization, freshness, and safety.

4. **Filtering Layer**

Removes unsafe, offensive, biased, spammy, or policy-violating suggestions.

5. **Delivery Layer**

Returns the top suggestions to the user interface with strict latency budgets.

3. Candidate Generation

Candidate generation must be extremely fast. The system should avoid running a large model for every keystroke whenever possible.

For autocomplete, one of the most effective structures is a **prefix trie**. A trie stores words or phrases in a tree-like structure where each path represents a prefix. This allows fast lookup for completions such as:

User types: mach

Suggestions: machine, machine learning, machine translation

Other candidate sources include:

- ↳ Static dictionary terms.
- ↳ Frequently used user phrases.
- ↳ App-specific vocabulary.
- ↳ Contact names or entity names.
- ↳ N-gram predictions.
- ↳ Small on-device neural language models.
- ↳ Server-side transformer models for richer suggestions.

For spell-check, candidate generation may use edit distance, phonetic similarity, keyboard proximity, and confusion sets. For example, *recieve* may generate *receive*, while *teh* may generate *the*.

For smart replies, candidate generation can use a small set of high-quality templates or a compact generative model. In sensitive applications like email, generated replies should be short, conservative, and easy to review.

4. Ranking and Personalization

After candidates are generated, the ranking model decides which suggestions appear first.

The ranking model may use features such as:

- ↳ Current prefix.
- ↳ Previous words and sentence context.
- ↳ User's historical acceptance rate.
- ↳ Time of day and app context.
- ↳ Language and locale.
- ↳ Domain-specific terminology.
- ↳ Recent repeated phrases.
- ↳ Candidate length.
- ↳ Model confidence.
- ↳ Safety score.

A lightweight neural ranker or gradient-boosted model can be used when latency is strict. For richer contexts, a transformer-based model may rerank candidates server-side, but only if the product can tolerate additional latency.

Personalization is valuable because typing behavior is highly individual. One user may often type technical terms like “embedding,” “retrieval,” or “latency,” while another may use legal, medical, or sales vocabulary. However, personalization must be designed carefully because typed text can contain passwords, health information, financial data, or private messages.

A strong production design uses **on-device personalization** whenever possible. The system can learn frequent phrases locally without uploading raw text. Federated learning can improve global models by sending encrypted model updates rather than raw user data.

5. Privacy and Security Design

Typing assistance systems process extremely sensitive data. Privacy is not optional; it is a core system requirement.

A strong privacy-first design includes:

- ↳ On-device inference for common predictions.
- ↳ No logging of raw sensitive text by default.
- ↳ Local personalization stored securely on the device.
- ↳ Differential privacy for aggregated analytics.
- ↳ Federated learning for model improvement.
- ↳ Data minimization and short retention windows.
- ↳ Redaction of passwords, credit cards, authentication codes, and private identifiers.

The system should also detect sensitive fields. For example, suggestions should be disabled or heavily restricted in password fields, payment fields, medical forms, and authentication-code inputs.

Security controls should include encryption in transit, encryption at rest, strict access controls, audit logs, and abuse monitoring. In an interview, it is important to emphasize that the system should not blindly send every keystroke to the server.

6. Safety and Quality Controls

Typing suggestions can cause user harm if they are offensive, biased, misleading, or inappropriate for the context. Safety filtering should happen before suggestions are displayed.

The system should block or downrank:

- ↳ Hate speech.
- ↳ Harassment.
- ↳ Sexual or violent content in inappropriate contexts.
- ↳ Sensitive personal data leakage.
- ↳ Spammy or manipulative phrases.
- ↳ Unsafe medical, legal, or financial advice.
- ↳ Suggestions that impersonate the user too aggressively.

For smart replies, the system should prefer neutral, low-risk responses. For example, in workplace email, suggestions like “Sounds good,” “Thanks for the update,” or “I’ll review this” are safer than overly emotional or legally sensitive responses.

Human evaluation is also important. Offline benchmarks are useful, but typing assistance is deeply user-experience driven. The best system is not always the one with the highest language-model probability; it is the one users trust and accept.

7. Latency and Scaling Strategy

Latency is one of the hardest parts of the design. Since suggestions may be requested after every keystroke, the system must avoid unnecessary computation.

Optimization strategies include:

- ↳ On-device inference for first-pass suggestions.
- ↳ Prefix trie lookup for instant candidate generation.
- ↳ Caching frequent prefixes and completions.
- ↳ Debouncing requests to avoid scoring every keystroke.
- ↳ Small distilled models for mobile devices.
- ↳ Quantization to reduce model size.

- ↳ Early exit ranking when confidence is high.
- ↳ Server fallback only for complex contexts.

A practical latency budget might look like:

- ↳ Input preprocessing: 2–5 ms.
- ↳ Candidate generation: 5–10 ms.
- ↳ Ranking: 10–20 ms.
- ↳ Safety filtering: 5–10 ms.
- ↳ UI rendering: 5 ms.

Total target: under 50 ms for common predictions.

For large-scale server-side systems, use regional inference endpoints, load balancing, autoscaling, model caching, and request batching where possible. However, batching must be carefully managed because typing interactions are latency-sensitive.

8. Metrics and Evaluation

Typing assistance should be evaluated using both ML metrics and product metrics.

Important offline metrics include:

- ↳ Top-k accuracy.
- ↳ Mean reciprocal rank.
- ↳ Word error correction rate.
- ↳ False correction rate.
- ↳ Toxic suggestion rate.
- ↳ Latency percentiles: p50, p95, p99.

Important product metrics include:

- ↳ Keystrokes saved.
- ↳ Suggestion acceptance rate.
- ↳ Edit-after-acceptance rate.
- ↳ User disable rate.
- ↳ Retention and engagement.
- ↳ False positive spell-correction complaints.
- ↳ Smart reply usage rate.

The most important metric is often **keystrokes saved without reducing user trust**. A suggestion that saves time but feels intrusive, wrong, or unsafe can damage the product experience.

9. Interview Trade-Offs to Discuss

A strong candidate should clearly explain trade-offs.

On-device vs server-side

On-device inference improves privacy and latency, but server-side models may be more accurate and easier to update.

Generative vs retrieval-based suggestions

Generative models can produce flexible suggestions, but retrieval-based systems are faster, safer, and easier to control.

Personalization vs privacy

Personalization improves relevance, but raw user text should not be uploaded or stored unnecessarily.

Accuracy vs latency

A larger model may improve suggestion quality, but typing assistance requires instant response.

Aggressive correction vs user control

Spell-check should help, not override. Incorrect autocorrection can be

worse than no correction.